

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

School of Computing — Department of Computer Science & Engineering

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Course: 21CSC203P — Advanced Programming Practice	Department: Computer Science and Engineering

Unit III | Session 3 | SLO 2: Callable and Future (Code: 3032)

Q: 1. Fill in the blank:

Unlike Runnable, the _____ interface can return a computed result.

Ans: Answer: Callable (java.util.concurrent.Callable)

Q: 2. True or False:

Callable's call() method can throw checked exceptions.

Ans: Answer: True

Q: 3. Multiple Choice:

What method does the Callable interface define?

A) run() B) execute() C) call() D) compute()

Ans: Answer: C) call()

Q: 4. Fill in the blank:

A _____ object represents the pending result of an asynchronous computation.

Ans: Answer: Future

Q: 5. What does the Future.get() method do if the task is still calculating?

Ans: Answer: It blocks the calling thread until computation finishes.

Q: 6. True or False:

Future.isDone() returns true if the task completed normally, threw an exception, or was cancelled.

Ans: Answer: True

Q: 7. Multiple Choice:

Which method cancels task execution?

A) stop() B) abort() C) cancel() D) terminate()

Ans: Answer: C) cancel()

Q: 8. Fill in the blank:

Future.get(timeout, unit) throws _____ if the timeout expires before completion.

Ans: Answer: TimeoutException

Q: 9. Identify Error: Callable c = () -> { return 5; }; Thread t = new Thread(c);

Ans: Answer: Thread constructor accepts Runnable, not Callable directly (use FutureTask or ExecutorService).

Q: 10. Fill in the blank:

Callable is a parameterized functional interface with generic type _____.

Ans: Answer: Callable<V> (or <T>)